Lab Exercise 15- Creating ConfigMaps in Kubernetes

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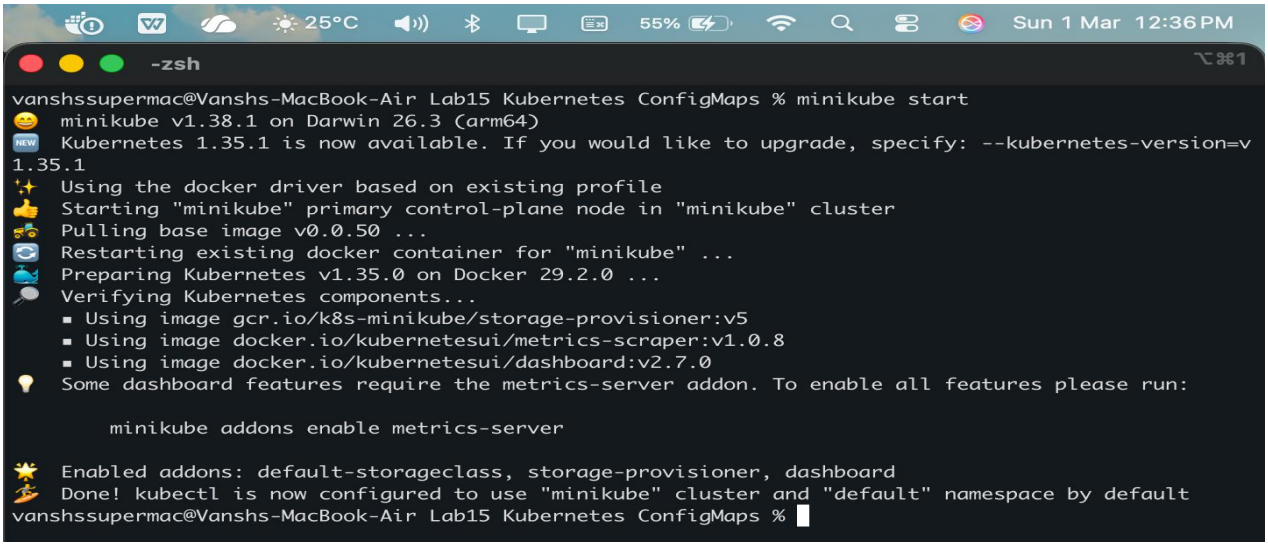
In this lab exercise, you will learn how to create ConfigMaps in Kubernetes using different methods: from literal values, from a file, from an environment file, and from multiple files. ConfigMaps allow you to decouple configuration artifacts from image content to keep containerized applications portable.

Step 1: Set Up Kubernetes Cluster

Ensure you have access to a Kubernetes cluster. You can use a local setup with Minikube, kind, or use a cloud-based Kubernetes service.



Step 2: Create a ConfigMap from Literal Values

Create a ConfigMap named example-configmap with literal values:

```
kubectl create configmap example-configmap --from-literal=key1=value1 --from-literal=key2=value2
```

```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl create configmap vb-configmap
 --from-literal=key1=value1 --from-literal=key2=value2
configmap/vb-configmap created
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps %
```

Verify the ConfigMap:

```
kubectl get configmap example-configmap -o yaml
```

```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl get configmap vb-configmap -o
 yaml
apiVersion: v1
data:
  key1: value1
  key2: value2
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-01T07:11:31Z"
  name: vb-configmap
  namespace: default
  resourceVersion: "7420"
  uid: ee30bad0-37fb-41f3-a967-7302af37f651
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps %
```

You should see the example-configmap with the keys and values specified.

Step 3: Create a ConfigMap from a File

Create a file named **config.txt** with the following content:

key1=value1

key2=value2

```
key1=value1
key2=value2
~
~
```

Create a ConfigMap named file-configmap from the file:

kubectl create configmap file-configmap --from-file=config.txt

```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % touch config.txt
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % vi config.txt
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl create configmap file-configm
ap --from-file=config.txt
configmap/file-configmap created
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps %
```

Verify the ConfigMap:

kubectl get configmap file-configmap -o yaml

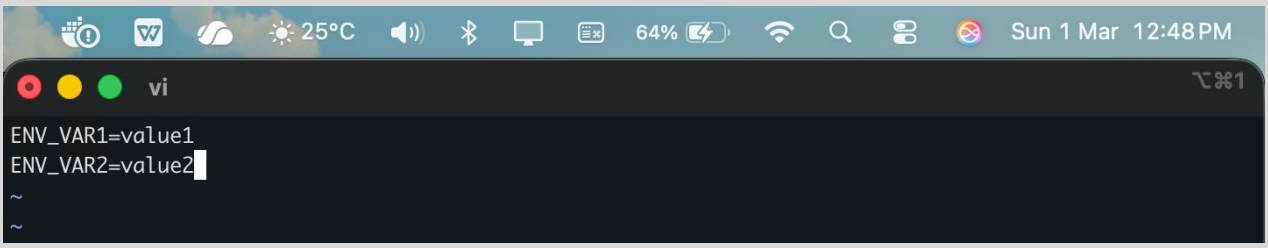
```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl get configmap file-configmap
-o yaml
apiVersion: v1
data:
  config.txt: |
    key1=value1
    key2=value2
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-01T07:15:41Z"
  name: file-configmap
  namespace: default
  resourceVersion: "7620"
  uid: 40fe0377-e6a1-4797-b6d8-59a6a970e66f
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps %
```

You should see the file-configmap with the content of config.txt.

Step 4: Create a ConfigMap from an Environment File

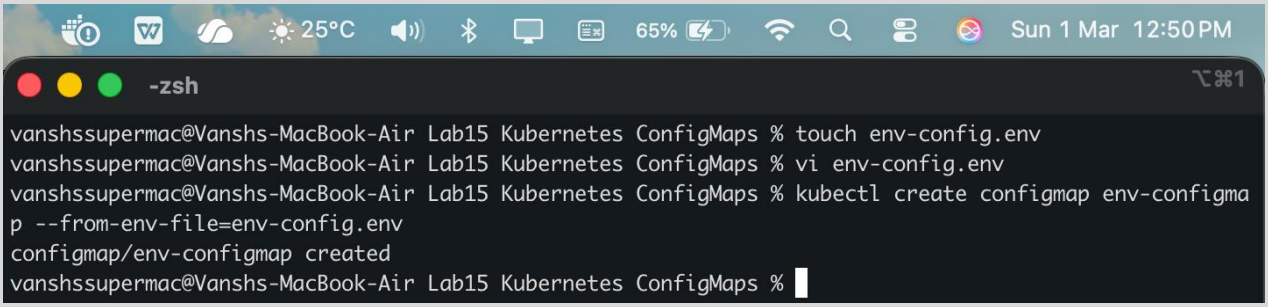
Create an environment file named **env-config.env** with the following content:

```
ENV_VAR1=value1
ENV_VAR2=value2
```



Create a ConfigMap named env-configmap from the environment file:

```
kubectl create configmap env-configmap --from-env-file=env-config.env
```



Verify the ConfigMap:

```
kubectl get configmap env-configmap -o yaml
```

```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl get configmap env-configmap -
o yaml
apiVersion: v1
data:
  ENV_VAR1: value1
  ENV_VAR2: value2
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-01T07:19:50Z"
  name: env-configmap
  namespace: default
  resourceVersion: "7820"
  uid: c2c3cd06-5c72-4f01-89ab-623d2f9bca62
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps %
```

You should see the env-configmap with the environment variables specified in the file.

Step 5: Create a ConfigMap from Multiple Files

Create multiple files:

config1.txt with content:

```
key1=value1
```

```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % touch config1.txt
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % vi config1.txt
```

config2.txt with content:

```
key2=value2
```

```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % touch config2.txt
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % vi config2.txt
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps %
```

Create a ConfigMap named multi-file-configmap from multiple files:

kubectl create configmap multi-file-configmap --from-file=config1.txt --from-file=config2.txt

```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl create configmap vb-ultimate
--from-file=config1.txt --from-file=config2.txt
configmap/vb-ultimate created
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps %
```

Verify the ConfigMap:

kubectl get configmap multi-file-configmap -o yaml

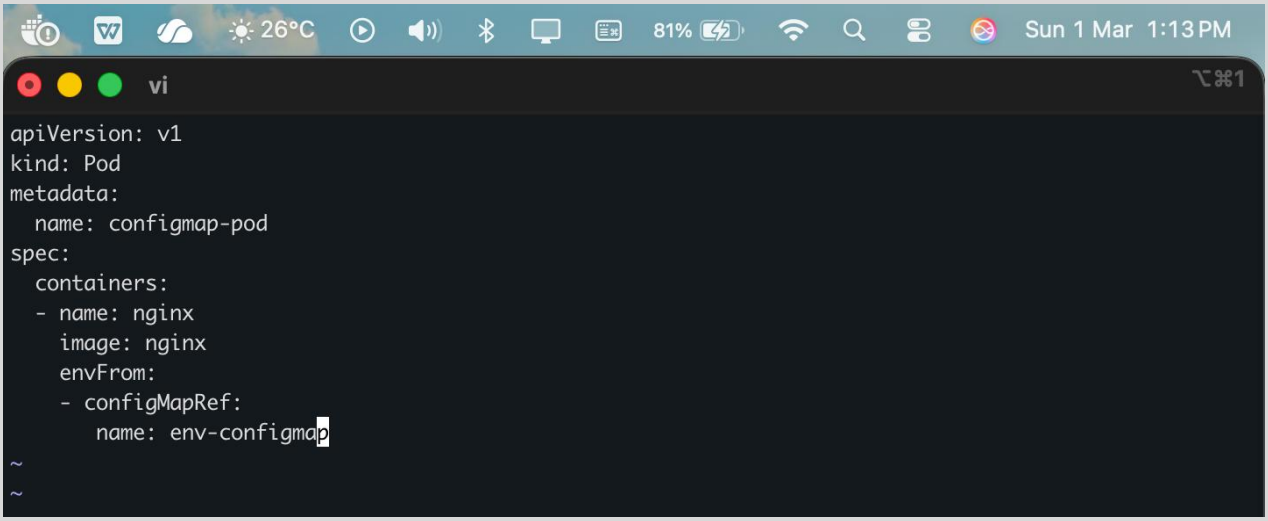
```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl get configmap vb-ultimate -o
yaml
apiVersion: v1
data:
  config1.txt: |
    key1=value1
  config2.txt: |
    key2=value2
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-01T07:25:04Z"
  name: vb-ultimate
  namespace: default
  resourceVersion: "8071"
  uid: 765e3c4b-b876-4fc0-8d7d-d58d3bad296d
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps %
```

You should see the multi-file-configmap with the content of both config1.txt and config2.txt.

Step 6: Use ConfigMaps in a Pod

Create a file named **pod-configmap.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: configmap-pod
spec:
  containers:
  - name: nginx
    image: nginx
    envFrom:
    - configMapRef:
        name: env-configmap
```



In this manifest:

- The Pod reads environment variables from the env-configmap.
- The Pod mounts the file-configmap at /etc/config.

Apply the manifest to create the Pod:

```
kubectl apply -f pod-configmap.yaml
```

```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % vi pod-configmap.yaml
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl apply -f pod-configmap.yaml
pod/configmap-pod created
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps %
```

Check the status of the Pod:

kubectl get pods

```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl get pods
NAME            READY   STATUS    RESTARTS   AGE
configmap-pod   1/1     Running   0          23s
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps %
```

View the Pod logs to see the environment variables and file content:

kubectl exec -it configmap-pod -- printenv

```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl exec -it configmap-pod -- pri
ntenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=configmap-pod
TERM=xterm
ENV_VAR2=value2
ENV_VAR1=value1
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
NGINX_VERSION=1.29.5
NJS_VERSION=0.9.5
```

You should see the environment variables and the content of the file-configmap.

Step 7: Clean Up

After completing the exercise, clean up the resources created:

kubectl delete pod configmap-pod

```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl delete pod configmap-pod
pod "configmap-pod" deleted from default namespace
```

kubectl delete configmap example-configmap

kubectl delete configmap file-configmap

kubectl delete configmap env-configmap

kubectl delete configmap multi-file-configmap

rm config.txt env-config.env config1.txt config2.txt

```
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl delete configmap vb-configmap

configmap "vb-configmap" deleted from default namespace
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl delete configmap file-configm
ap
configmap "file-configmap" deleted from default namespace
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl delete configmap env-configma
p
configmap "env-configmap" deleted from default namespace
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl delete configmap vb-utltimate

Error from server (NotFound): configmaps "vb-utltimate" not found
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % kubectl delete configmap vb-ultimate
configmap "vb-ultimate" deleted from default namespace
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps % rm config.txt env-config.env config1.
txt config2.txt
vanshssupermac@Vanshs-MacBook-Air Lab15 Kubernetes ConfigMaps %
```

Thank You